

15.2 Solution: By Eq. (15.4), the cross section for scattering and at the same time for producing a soft photon of energy $\hbar\omega$, per unit energy interval and per unit solid angle is

$$\frac{d^3\sigma}{d(\hbar\omega)d\Omega d\Omega_p} = \frac{d^2N}{d(\hbar\omega)d\Omega} \cdot \frac{d\sigma}{d\Omega_p}.$$

For classical elastic hard sphere scattering, the cross section is

$$\frac{d\sigma}{d\Omega_p} = \frac{1}{4}R^2,$$

and the differential number spectrum per unit energy interval and per unit solid angle of soft photons, in the dipole approximation and summed over incoming polarization, is given by Eq. (15.7),

$$\frac{d^2N}{d(\hbar\omega)d\Omega} = \frac{e^2}{4\hbar\pi^2c} \frac{1}{\hbar\omega} \sum_{i=1}^2 |\epsilon_i^* \cdot \Delta\beta|^2,$$

with $\Delta\beta = \beta_f - \beta_i$.

Assume that the particle initially travels along the z -direction, after scattering, its direction becomes (θ_p, ϕ_p) in spherical coordinates, *i.e.*,

$$\beta_i = \beta(0, 0, 1), \quad \beta_f = \beta(\sin\theta_p \cos\phi_p, \sin\theta_p \sin\phi_p, \cos\theta_p).$$

Therefore,

$$\Delta\beta = \beta(\sin\theta_p \cos\phi_p, \sin\theta_p \sin\phi_p, \cos\theta_p - 1).$$

Notice that elastic scattering does not change the magnitude of the velocity, but only its direction. Also,

$$\mathbf{n} = (\sin\theta, 0, \cos\theta), \quad \epsilon_1 = (\cos\theta, 0, -\sin\theta), \quad \epsilon_2 = (0, 1, 0),$$

we have

$$\epsilon_1^* \cdot \Delta\beta = \beta(\cos\theta \sin\theta_p \cos\phi_p - \sin\theta(\cos\theta_p - 1)), \quad \epsilon_2^* \cdot \Delta\beta = \beta \sin\theta_p \sin\phi_p.$$

Therefore,

$$\frac{d^3\sigma}{d(\hbar\omega)d\Omega d\Omega_p} = \frac{e^2}{4\pi^2c} \frac{\beta^2}{\hbar^2\omega} \frac{R^2}{4} \left[(\cos\theta \sin\theta_p \cos\phi_p - \sin\theta(\cos\theta_p - 1))^2 + \sin^2\theta_p \sin^2\phi_p \right].$$

The differential cross section can be obtained by integrating with respect to the solid angle of classical elastic scattering,

$$\begin{aligned} \frac{d^2\sigma}{d(\hbar\omega)d\Omega} &= \int \frac{d^3\sigma}{d(\hbar\omega)d\Omega d\Omega_p} d\Omega_p \\ &= \frac{e^2}{4\pi^2c} \frac{\beta^2}{\hbar^2\omega} \frac{R^2}{4} \int \left[\cos^2\theta \sin^2\theta_p \cos^2\phi_p + \sin^2\theta(1 + \cos^2\theta_p) + \sin^2\theta_p \sin^2\phi_p \right] d\Omega_p, \end{aligned}$$

where we have dropped those terms that will be zero upon integration. Finally, notice that

$$\int d\Omega_p = 4\pi, \quad \int \cos^2\theta_p d\Omega_p = \int \sin^2\theta_p \cos^2\phi_p d\Omega_p = \int \sin^2\theta_p \sin^2\phi_p d\Omega_p = \frac{4\pi}{3},$$

we will have

$$\begin{aligned} \frac{d^2\sigma}{d(\hbar\omega)d\Omega} &= \frac{e^2}{4\pi^2c} \frac{\beta^2}{\hbar^2\omega} \frac{R^2}{4} \left[\cos^2\theta \cdot \frac{4\pi}{3} + \sin^2\theta \left(\frac{4\pi}{3} + 4\pi \right) + \frac{4\pi}{3} \right] \\ &= \frac{R^2}{16\pi^2} \frac{e^2}{\hbar c} \frac{\beta^2}{\hbar\omega} \left(\frac{8\pi}{3} + 4\pi \sin^2\theta \right) = \frac{R^2}{12\pi} \frac{e^2}{\hbar c} \frac{\beta^2}{\hbar\omega} (2 + 3\sin^2\theta). \end{aligned}$$